**NYU CS-GY 6923: Machine Learning** Fall 2024

Homework 1

Name: Star Student

# Problem 1

**Collaborators:** List all collaborators here, or simply write "None" if you worked alone.

Problems will be answered with a combination of written explanations, equations, plots, and images.

$$\begin{aligned}\mathbb{E}[X+Y] &= \mathbb{E}[X] + \mathbb{E}[Y] \\ &= 42.\end{aligned}$$

If you reference any outside material or theorems not discussed in class, please add a link and explanation: https://xkcd.com/2042/.



Figure 1: Please explain all plots or diagrams with a sufficiently detailed caption!

Occasionally you may be asked to include code snippets:

```
def majorityElement(self, nums):
count = 0
candidate = None

for num in nums:
if count == 0:
candidate = num
count += (1 if num == candidate else -1)

return candidate
```

And sometimes it might be helpful to include pseudocode:

$count \leftarrow 0$
$candidate \leftarrow null$
**for** $i \in a_1, \ldots, a_n$ **do**
    **if** $count == 0$ **then**
        $candidate \leftarrow a_i$
    **end if**
    $count \leftarrow count + 1$
**end for**
**return** *candidate*

## Problem 2

**Collaborators:** List all collaborators here, or simply write "None" if you worked alone.

Please start each new problem on a new page. This makes it easier to separate out problem sets during grading.